ISOMORPHISMS APPEAR AT A FINITE STAGE IN THE DESCENT SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1934-1943, following the statement at 1924-1932.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Iso = etale + universally injective + surjective. Morphisms of Spaces, Lemma spaces-morphisms-lemma-etale-universally-injective-open (cross-chapter, cited at ~1936-1940). Decomposes the conclusion into three descent properties.
- Three descent engines. lemma-descend-etale (same chapter, card L26333); lemma-descend-surjective (same chapter, card L26329); lemma-descend-universally-injective (same chapter, card L26359). Each component appears at a finite stage; take the maximum index.

Source reference. Original lemma 07SL: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1677–1677. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07SN: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1776–1776. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 084X: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1826–1826. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1666–1666. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05W5: focused-seeds/geometry/S9-spaces-limits-descend-isomorphism/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction

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the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Notation and assumptions as in Situation 084W. If*

(1) *f is an isomorphism,*
(2) *f_0 is locally of finite presentation,*

then f_i is an isomorphism for some $i \geq 0$.

Proof. Being an isomorphism is equivalent to being étale, universally injective, and surjective, see Morphisms of Spaces, Lemma 05W5. Thus the lemma follows from Lemmas 07SL, 07SN, and 084X. □